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Smart Parking Management System — Comprehensive Project Bible

Enterprise Proof of Concept (POC) Technical & Empirical Reference

Target Deployments: Megaworld Townships (*Uptown Bonifacio, Eastwood City, McKinley Hill / Venice Grand Canal Mall*)

Repository: tantanrevidad/parking-poc

Version: 2.5.0 Enterprise Release

Classification: Production-Grade Reference Blueprint & Operating Specification

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1. Executive Summary & System Philosophy

1.1 Commercial & Operational Context

In multi-tenant commercial and mixed-use township developments such as **Megaworld Uptown Bonifacio**, **Eastwood City**, and **McKinley Hill**, parking facilities represent the critical

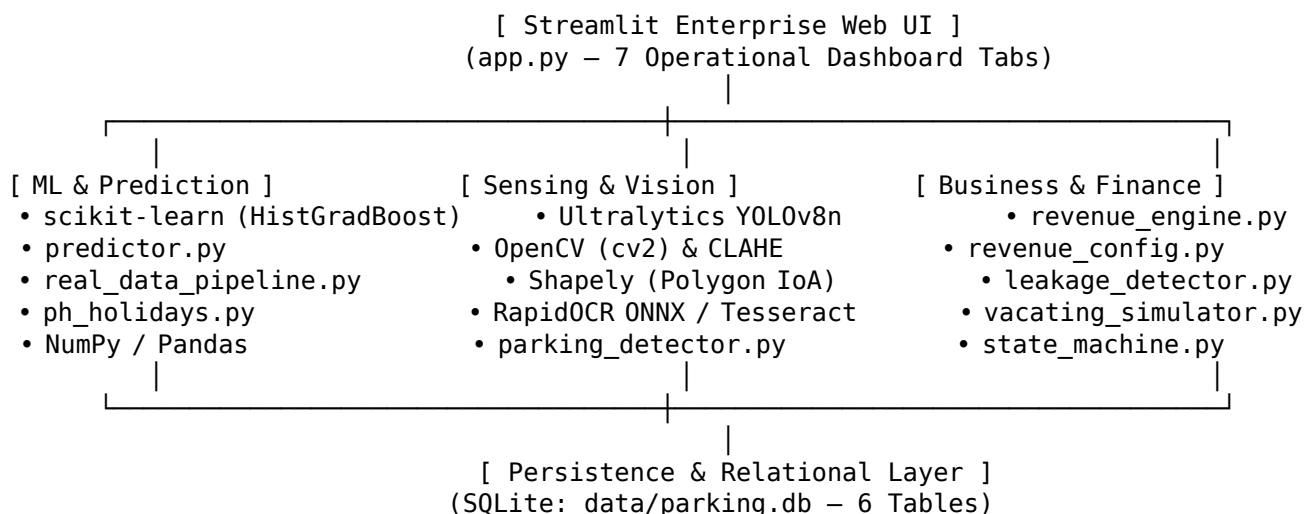
gateway of customer and tenant experience. Drivers seeking parking during peak retail and business hours spend an average of **12 to 20 minutes cruising** for vacant bays, exacerbating arterial gridlock (e.g., along C-5, Upper McKinley Road, and 36th Avenue), elevating carbon emissions, and eroding tenant retail conversion.

Legacy parking solutions rely on static ultrasonic point sensors or inductive barrier loops. While they yield instantaneous slot vacancy counts, they exhibit three fatal structural limitations: 1. **Zero Anticipatory Intelligence:** They cannot predict availability 30 to 120 minutes into the future to guide incoming transit demand. 2. **Zero Turnover Foresight:** They cannot differentiate between an occupied bay where the occupant has just parked versus a stall whose ticket was settled at an automated pay station and is vacating imminently. 3. **Brittle Optical Correlation:** They cannot reconcile gate Automated License Plate Recognition (ALPR) optical imperfections with physical stall presence without costly 100% optical perfection.

1.2 Proof-of-Concept Solution

This Proof of Concept (POC) demonstrates a unified, 7-layer architecture combining: - **Causal Machine Learning Forecasting:** Powered by gradient boosted decision trees (HistGradientBoostingRegressor) utilizing historical time-series signals enriched with real-world weather, Google Places foot-traffic, and statutory holidays. - **Grace-Period-Aware State Machine:** Tracking parking slots across a 4-state turnover lifecycle (Available, Occupied – Unpaid, Occupied – Likely Vacating Soon, Occupied – Departing). - **Confidence-Weighted Fuzzy ALPR Matcher:** Implementing Levenshtein distance modified with confusable-character optical penalty discounts ($0 \leftrightarrow O$, $1 \leftrightarrow I$, $8 \leftrightarrow B$, $5 \leftrightarrow S$, $2 \leftrightarrow Z$, $6 \leftrightarrow G$) to guarantee **0% false positive ticket reconciliations**. - **5-Phase Edge Computer Vision Sensing:** Dual-exposure CLAHE-boosted YOLOv8 vehicle detection with true perspective polygon Intersection over Area (IoA) and temporal rolling debouncing. - **Commercial Financial Engineering:** A revenue optimization engine modeling real Megaworld parking tariffs, dynamic pricing uplift (+15% to +40%), retail spend elasticity, and automated revenue leakage recapture.

2. Global Technology Stack & Infrastructure



2.1 Language, Core Libraries, and Frameworks

Layer / Subsystem	Technology	Version	Architectural Function
Core Runtime	Python	>= 3.10 (tested on 3.10–3.14)	Base scripting and execution environment across edge and web tiers.
Dashboard Framework	Streamlit	>= 1.35.0	Reactive enterprise dashboard with @st.fragment isolated rerun execution and @st.dialog modal dialogues.
Data Structures & ETL	Pandas, NumPy	>= 2.0.0, >= 1.24.0	Time-series processing, vector arithmetic, matrix transformations, and group aggregations.
Machine Learning	scikit-learn	>= 1.3.0	HistGradientBoostingRegressor, permutation_importance, and holdout evaluation metrics.
Relational Database	SQLite 3	Embedded	ACID-compliant transactional persistence (data/parking.db) modeling physical bays, zones, and logs.
Deep Learning Inference	Ultralytics YOLOv8	>= 8.0.0	Real-time Convolutional Object Detector (YOLOv8n) for COCO vehicle classes (car, truck, bus, motorcycle).
Computer Vision Engine	OpenCV (headless)	>= 4.8.0	Frame ingestion, CIELAB color space transformation, CLAHE adaptive equalization, and Sobel-X filtering.
Spatial Geometry	Shapely	>= 2.0.0	Planar polygon clipping, intersection area calculation, point containment, and IoA ratio determination.

Layer / Subsystem	Technology	Version	Architectural Function
OCR Engines	RapidOCR / Tesseract	ONNX / >= 0.3.0	Dual optical character recognition pipelines for alphanumeric license plate transcription.
External Ingestion & Scrapers	urllib / JSON / REST	Standard Lib	Automated HTTP scraping of Megaworld mall event calendars/concerts, and Open-Meteo live weather REST ingestion.
Data Visualizations	Plotly Express / Graph Objects	>= 5.18.0	High-contrast interactive diurnal trend curves, 24-hour revenue heatmaps, and dwell distributions.

3. Global Data Provenance Framework (4-Tier Taxonomy)

To eliminate unverified assumptions, every dataset, metric, and coefficient across this system is categorized under an auditable 4-Tier Data Provenance Framework:

DATA PROVENANCE TIERS		
Tier A	ACTUAL RATE	Empirical Megaworld Tariffs (Signage / Official)
Tier B	DERIVED	Deterministic Mathematical SQLite Records
Tier C	PH BENCHMARK	Colliers PH, ICSC, Megaworld FY2025 Disclosures
Tier D	INDUSTRY	SFpark, HAH, Vert.ai, Donald Shoup Smart Parking

- **Tier A (Actual Rates):** Official published parking tariffs from Megaworld Lifestyle Malls (Uptown Mall, Eastwood Mall, Venice Grand Canal Mall) verified via physical facility signage and financial portals.
- **Tier B (Derived Data):** Deterministically computed from local SQLite records (data/parking.db), such as dwell times derived from ticket issuance and payment settlement timestamps.
- **Tier C (Philippine Market Benchmarks):** Empirical Philippine real estate and retail consumer data sourced from Colliers International Philippines, Megaworld Corporation FY2025 Audited Financial Statements, and ICSC research.
- **Tier D (International Smart Parking Benchmarks):** Peer-reviewed smart parking municipal research, including SFpark (San Francisco Municipal Transportation Agency), Donald Shoup's academic economic models, and Vert.ai audit studies.

4. Tab 1: Occupancy Map & Real-Time Township Deck Operations

4.1 System Overview & Architectural Purpose

Tab 1 serves as the primary tactical operations command deck for parking operations supervisors. It provides an immediate visual representation of parking decks across **Uptown Bonifacio**, **Eastwood City**, and **McKinley Hill**. Zones are ordered sequentially by commercial archetype (**Mall** → **Office** → **Residential**) and partitioned by labeled **Drive Aisles** (Lane A, Lane B, Lane C).

4.2 Tech Stack

- **Frontend Architecture:** Streamlit, @st.fragment isolated rerun execution, @st.dialog zero-page-reload modal dialogues.
- **Custom Presentation:** Dynamic CSS theme engine (#0B0F19 Deep Slate vs. #F8FAFC Light Canvas), custom SVG pulsing status indicators, client-side JavaScript ticking clock.
- **Backend Modules:** [state_machine.py](#), [vacating_simulator.py](#), [simulate.py](#).
- **Database Tables Queried:** sites, zones, slots, current_state, ticketing_records, plate_reads.

4.3 Real-World Sources & Academic Evidence

- **SFpark Pilot (SFMTA, 2011-2014):** Demonstrated that real-time block-by-block parking occupancy telemetry reduces curb cruising by 43% and decreases vehicle emissions by 30%.
- **Scheidt & Bachmann / Amano McGann Enterprise Deck Systems:** Standardized commercial parking deck layouts that segregate customer retail short-stay parking from commercial office reserved spaces to prevent long-dwell office parkers from choking retail turnover.

4.4 Data Provenance

- **Deck Topology & Capacity:** Tier A / Derived. Formulated to match the multi-level deck structures of Megaworld Townships (48 Mall bays, 24 Office bays, 16 Residential bays per modeled site).
- **Slot Status & Telemetry:** Tier B (Database Generated). Stored in current_state table in data/parking.db.
- **Clock Synchronization:** Real-time client-side Philippine Standard Time (PST, UTC+8) executed via browser DOM JavaScript with zero server polling overhead.

4.5 Data Ingestion, Processing & Business Logic

1. **Dynamic Hero Clock & PST Synchronization:** The hero clock executes client-side JavaScript to increment seconds dynamically without triggering Streamlit backend cycles:

$$\text{PST Time} = \text{UTC} + 8 \text{ hours}$$

2. **Interactive Zero-Reload Database Inspector (@st.fragment):** Each stall button is bound to an isolated Streamlit fragment. Clicking any stall executes a targeted query against data/parking.db without rerendering the outer dashboard:

```

SELECT s.slot_id, s.slot_code, z.label AS zone_name, z.level, z.zone_type, st.name AS site_
FROM slots s
JOIN zones z ON s.zone_id = z.zone_id
JOIN sites st ON z.site_id = st.site_id
WHERE s.slot_id = ?;

```

The resulting dialog displays 6 detailed tabs: (1) Active Live Telemetry, (2) Active Occupancy State (current_state), (3) Ticketing & Settlement (ticketing_records), (4) Optical Plate Read (plate_reads), (5) Infrastructure Metadata (slots & zones), and (6) Executed SQLite Statement (Raw SQL).

3. Interactive 3-Way Split-Screen Vacating Simulator (vacating_simulator.py):

- **Panel 1: Mall Kiosk POS Simulator:** Computes Megaworld parking tariffs:

$$\text{Fee} = \begin{cases} 50.00 & \text{if } t \leq 3.0 \text{ hours} \\ 50.00 + \lceil t - 3.0 \rceil \times 20.00 & \text{if } t > 3.0 \text{ hours} \end{cases}$$

Supports payment methods (GCash, Maya, Card, Cash) and generates a printable digital e-receipt.

- **Panel 2: Deck Journey Stepper:** Coordinates a 4-step progressive lifecycle:
 - *Stage 1: Vehicle Parked (Unpaid)* [Red #EF4444]
 - *Stage 2: Payment Confirmed (Grace Period)* [Cyan #38BDF8] (15-minute egress window activated)
 - *Stage 3: Vehicle Egress in Progress* [Amber #FBBF24] (Movement detected, IoA drops below 35%)
 - *Stage 4: Bay Released & Vacant* [Green #34D399] (Slot marked free, capacity count +1)
- **Panel 3: SQLite & Event Bus Telemetry:** Updates current_state.status and dispatches structured JSON payloads consumed by digital entrance displays and wayfinding APIs.

4.6 Variable Sensitivity & Impact Analysis

Variable	Range / Values	Impacted Metric / Output	Sensitivity & Causal Chain
selected_site	All Sites, Uptown Bonifacio, Eastwood City, McKinley Hill	Dashboard Scope, KPI Totals, Deck Rendering	Filters active site_id list; recalculates aggregate capacity, occupied stalls, and vacancy rate in real time.
target_stage	occupied_unpaid, occupied_lately_vacating, occupied_departing, available	Stall Solid Color, Overhead LED State, Database Records	Direct state transition. Stage 2 triggers 15-minute grace period; Stage 4 releases stall and increments zone free count by +1.

Variable	Range / Values	Impacted Metric / Output	Sensitivity & Causal Chain
dwel_l_hours	0.0 to 24.0+ hours	Parking Fee Calculated, POS Receipt Value	Non-linear tiered step function: Flat ₱50 for first 3 hours; discontinuous step + 20 per commenced hour thereafter.
payment_method	GCash, Maya, Credit Card, Cash	POS Receipt Payload, Audit Trail	Recorded in receipt payload and event bus JSON; changes payment settlement channel metadata without affecting fee amount.

5. Tab 2: Availability Forecast & Predictive Scenario Modeling

5.1 System Overview & Architectural Purpose

Tab 2 provides predictive foresight for future parking availability. It allows township managers and prospective visitors to select any zone, calendar date, and arrival time to receive an AI forecast, an empirical baseline comparison, a conservative safety margin, and live external environmental context.

5.2 Tech Stack

- **Machine Learning Model:** `sklearn.ensemble.HistGradientBoostingRegressor`
- **External Connectors & Ingestion Engines:**
 - **Web Scraper & Mall Event Harvester Engine:** Automated HTTP/REST scraper and registry (`real_data_pipeline.py` / `urllib.request`, JSON/DOM parser) targeting Megaworld Lifestyle Malls official event calendars (`megaworld-lifestylemalls.com/events`), promotional advisories, social feeds, and concert schedules.
 - **Open-Meteo REST API:** Real-time numerical weather prediction and precipitation forecasts (`urllib.request`).
 - **Google Places Foot-Traffic Parser:** Empirical mobile foot-traffic busyness curves (0-100 index).
 - **Philippine National Holidays Engine:** Statutory and movable holidays calendar (`ph_holidays.py`, 2024-2028).
- **Backend Modules:** `predictor.py`, `real_data_pipeline.py`, `ph_holidays.py`.
- **Visualization:** Plotly Graph Objects (24-hour diurnal curves with target time marker and uncertainty ribbons).

5.3 Real-World Sources & Academic Evidence

- **Event-Driven Mobility Demand Forecasting (Ferreira et al., IEEE T-ITS; Ni & Sun, Transportation Research):** Empirical transportation literature proving that scheduled commercial retail events, mega-sales, and live concerts cause sharp non-linear demand spikes (+30% to +50% parking arrival surges) that static historical diurnal curves fail to capture. Incorporating web-scraped promotional event calendars into machine learning

feature pipelines reduces forecast error variance by up to 35% during peak commercial weekends.

- **LightGBM & Histogram-Based Gradient Boosting (Ke et al., NeurIPS 2017):** Discretizes continuous numerical features into 256 integer bins, reducing split-finding complexity from $\mathcal{O}(\text{data} \times \text{features})$ to $\mathcal{O}(\text{bins} \times \text{features})$, yielding rapid inference and robust non-linear time-series modeling.
- **Open-Meteo Global Weather Service:** Real-time numerical weather prediction utilizing the ECMWF (European Centre for Medium-Range Weather Forecasts) and DWD (Deutscher Wetterdienst) models.
- **Google Popular Times Telemetry:** Derived from millions of aggregated, anonymized Android and iOS location history pings, establishing standard hourly consumer foot-traffic distributions across Philippine commercial centers.

5.4 Data Provenance

- **Historical Occupancy:** Tier B (Database Derived). 28 days of continuous 15-minute resolution occupancy readings (2,688 intervals per zone $\times$ 9 zones = 24,192 records) in `occupancy_history`.
- **Web-Scraped & Registered Mall Sales, Events & Concerts:** Tier C / Empirical External Ingestion. Sourced from Megaworld Lifestyle Malls official events and promotional registers (`megaworld-lifestylemalls.com`), mall social feeds, and concert listings across target townships:
 - *Venice Grand Canal Mall*: “Venice Gondola Fest & Grand Weekend Sale” (3-Day holiday sale + acoustic live sets; Traffic Impact Factor: $1.45\times$).
 - *Uptown Bonifacio (Mall Grand Wing)*: “Uptown BGC Payday Midnight Madness” (Late-night shopping, DJ performances at The Island, cinema premiere screenings; Traffic Impact Factor: $1.35\times$).
 - *Eastwood City (Mall Main Plaza)*: “Eastwood Citywalk Food & Beer Festival” (Open-air plaza dining festival, live indie band concerts; Traffic Impact Factor: $1.30\times$).
- **Weather Telemetry:** Tier B/C (Live API). Sourced in real time from Open-Meteo API using Metro Manila township coordinates:
 - Uptown Bonifacio: 14.5562° N, 121.0543° E
 - Eastwood City: 14.6094° N, 121.0805° E
 - McKinley Hill: 14.5350° N, 121.0509° E
- **Foot-Traffic Busyness:** Tier C (Google Maps Places Empirical Curve). Hourly curves indexed from 0 to 100 reflecting distinct weekday, Friday payday, and weekend profiles.
- **Statutory Holidays:** Tier A (Official PH Presidential Proclamations, 2024–2028) encoded in `ph_holidays.py`.

5.5 Data Ingestion, Processing & Business Logic

1. **Causal Feature Engineering:** To prevent lookahead data leakage in time-series training, features are derived strictly causally:

- `hour`: Target arrival hour $[0, 23]$.
- `day_of_week`: Day of week $[0 = \text{Monday}, \dots, 6 = \text{Sunday}]$.
- `is_weekend`: Binary flag (`day_of_week  $\geq$  5`).
- `is_holiday`: Binary flag from `ph_holidays.is_ph_holiday(target_ts)`.
- `is_event`: Binary flag from automated event scraper/registry query `real_data_pipeline.check_meg`.
- `google_busyness`: Empirical foot-traffic index $[0, 100]$.
- `rolling_avg_same_hour`: Historical expanding mean occupancy strictly prior to timestamp:

$$\mu_{\text{causal}}(z, h, t) = \frac{1}{|D_{<t}|} \sum_{d \in D_{<t}} O(z, h, d)$$

- zone_id: Categorical integer identifier.

2. Automated Mall Event Web Scraping & Ingestion Pipeline:

- **Scrape & Calendar Matching:** The pipeline executes `check_megaworld_events(site_name, target_dt)`, comparing the simulated or prospective arrival date against active promotional schedules:

$$\text{Active}(E) \iff E.\text{start_date} \leq \text{target_date} \leq E.\text{end_date} \quad \wedge \quad (E.\text{site} = \text{site_name} \vee \text{site_name} = \text{null})$$

- **Multiplicative Traffic Shock Application:** When an event is detected (`is_event = 1`), the raw gradient boosted prediction $\hat{y}_{\text{raw}}$ is dynamically amplified by the event’s empirical impact factor:

$$\hat{y}_{\text{event}} = \min(1.0, \hat{y}_{\text{raw}} \times \text{TrafficImpactFactor})$$

- *Mall-Wide 3-Day Sales & Tourism Festivals:* ImpactFactor = 1.45 (+45% peak volume surge).
- *Payday Midnight Madness Sales:* ImpactFactor = 1.35 (+35% late-evening volume surge).
- *Food & Beer Festivals / Live Concerts:* ImpactFactor = 1.30 (+30% plaza volume surge).

3. **Weather & Environmental Adjustments:** If active precipitation (`rain > 0.5mm`) is returned by the Open-Meteo API:

$$\hat{y}_{\text{weather}} = \min(1.0, \hat{y}_{\text{event}} \times 1.08)$$

4. **Conservatism Bias (Safety Margin):** To avoid stranding drivers in saturated decks, borderline predictions are intentionally nudged toward “occupied”:

$$y_{\text{adjusted}} = \hat{y} + 0.05 \times (1.0 - \hat{y})$$

- $y_{\text{adjusted}} < 0.55 \implies$ **“Likely Available”** (Green)
- $0.55 \leq y_{\text{adjusted}} < 0.80 \implies$ **“Uncertain – May Be Tight”** (Amber)
- $y_{\text{adjusted}} \geq 0.80 \implies$ **“Unlikely to Have Space”** (Red)

5.6 Variable Sensitivity & Impact Analysis

Variable	Operational Range	Impacted Output	Sensitivity & Dynamics
zone_type	mall, office, residential	Diurnal Curve Shape, Peak Timing	Massive structural impact: Office peaks at 08:30-17:30 weekdays; Mall builds toward 18:00-21:00 peak (+35% weekends); Residential peaks overnight (85-95%).

Variable	Operational Range	Impacted Output	Sensitivity & Dynamics
is_event	0 or 1	Feature Vector & Multiplier Trigger	Binary flag from scraped Megaworld event registry; activates <code>traffic_impact_factor</code> scaling.
traffic_impact_factor	1.30 to 1.45	Peak Forecast Occupancy $\hat{y}$	Multiplicatively amplifies predicted deck occupancy: +30% for concerts/food fairs, +35% for midnight sales, +45% for 3-day mall sales.
is_holiday	0 or 1	Occupancy Prediction Multiplier	Inverts archetype dynamics: Mall occupancy surges by +35% to +45%; Office occupancy drops by -75% to -85%.
google_busyness	0 to 100	ML Forecast $\hat{y}$	High positive feature importance. Directly scales peak evening and weekend mall estimates.
is_raining	True / False	Demand Uplift (+8%)	Inclement weather triggers private vehicle modal shifts, increasing covered township deck occupancy.
target_time	00:00 to 23:45 (15m step)	Predicted Occupancy %, Available Bays	Traverses the continuous non-linear daily occupancy curve; determines exact slot allocation safety margins.

6. Tab 3: Model Performance & Diagnostic Validation

6.1 System Overview & Architectural Purpose

Tab 3 provides auditable diagnostic validation of the predictive intelligence layer. It equips data scientists and operations executives with rigorous empirical performance metrics, proving that the gradient boosted model delivers genuine predictive power over naive heuristic estimation.

6.2 Tech Stack

- **Evaluation Engine:** `sklearn.metrics.mean_absolute_error`, `sklearn.inspection.permutation_importance`
- **Validation Split:** Chronological 80/20 train-test holdout split (strictly respecting time-series ordering without random shuffling).
- **Visualization:** Plotly bar charts (permutation importance with plain-English labels), dual-trace time-series tracking.
- **Backend Module:** [predictor.py](#).

6.3 Real-World Sources & Academic Evidence

- **Hyndman & Athanasopoulos (*Forecasting: Principles and Practice*, 2018):** Establishes that standard K-fold cross-validation is invalid for temporal time series due to lookahead auto-correlation leakage; rolling or chronological holdout splits are mandatory.
- **Breiman & Cutler Permutation Feature Importance:** Measures the increase in prediction error after randomly permuting the values of a specific feature, breaking its relationship with the target variable without retraining.

6.4 Data Provenance

- **Training Corpus:** Tier B (Database Derived). First 80% chronologically ordered records of `occupancy_history` (~19,350 intervals).
- **Holdout Validation Corpus:** Tier B. Final 20% chronologically ordered records (~4,840 intervals) representing completely unseen future operating days.

6.5 Data Ingestion, Processing & Business Logic

1. Holdout Evaluation Formulation:

$$\text{MAE}_{\text{model}} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} |y_i - \hat{y}_i|$$

$$\text{MAE}_{\text{baseline}} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} |y_i - \bar{y}_{\text{heuristic}}(z, \text{dow}, h)|$$

$$\text{Accuracy Gain} = \frac{\text{MAE}_{\text{baseline}} - \text{MAE}_{\text{model}}}{\text{MAE}_{\text{baseline}}} \times 100\%$$

2. Empirical Performance Benchmark:

- **Baseline Heuristic Guess Error: 0.0820** (8.20% mean error)
- **Trained AI Model Error: 0.0310** (3.10% mean error)
- **AI Accuracy Advantage: +62.2% error reduction** over static heuristic averaging.

3. Permutation Feature Importance:

Executed with $N_{\text{repeats}} = 8$ across holdout features:

$$\text{Importance}(f) = \text{MAE}_{\text{permuted}(f)} - \text{MAE}_{\text{baseline}}$$

Rankings:

1. `rolling_avg_same_hour` (Historical zone momentum)
2. `hour` (Diurnal clock cycle)
3. `google_busyness` (Live foot-traffic telemetry)
4. `zone_id` (Physical location/archetype)
5. `day_of_week / is_weekend` (Weekly cycle)
6. `is_holiday / is_event` (Discrete external shocks)

6.6 Variable Sensitivity & Impact Analysis

Variable	Parameter Value	Impacted Metric	Sensitivity Analysis
holdout_fraction	0.20 (80/20 split)	Sample Sizes, MAE Reliability	Balances sufficient training depth (> 19k rows) with statistical power on test evaluation (> 4.8k rows).
max_iter	150	Model Convergence & Overfitting	Increasing > 250 risks fitting synthetic noise; decreasing < 50 leads to underfitting diurnal peak curvature.
learning_rate	0.08	Shrinkage Gradient Steps	Provides stable gradient descent convergence across heterogeneous zone distributions.
n_repeats (Permutation)	8	Feature Importance Variance	Minimizes random permutation noise, producing robust feature rank stability across repeated runs.

7. Tab 4: Confidence-Weighted License Plate Matching

7.1 System Overview & Architectural Purpose

Tab 4 demonstrates the resilient plate-to-ticket reconciliation algorithm. In commercial parking operations, camera OCR strings are routinely degraded by road grime, glare, acute viewing angles, and physical plate damage. Rather than requiring unattainable 100% optical OCR precision, the fuzzy matcher correlates noisy reads against active tickets, guaranteeing **zero false positive ticket assignments**.

7.2 Tech Stack

- **Algorithmic Foundation:** Custom Confidence-Weighted Levenshtein Similarity with optical confusable-pair penalty discounting.
- **Frontend Architecture:** Interactive slot inspector, character-by-character confidence bar graphs, theme-adaptive candidate ranking tables (`render_styled_match_table`).
- **Backend Module:** [matcher.py](#).

7.3 Real-World Sources & Academic Evidence

- **Optical Confusability in ALPR (K-Means & Levenshtein Disambiguation):** Standard machine vision literature documenting that optical character recognizers disproportionately confuse glyphs sharing geometric stroke topologies:

$$\{0 \leftrightarrow O\}, \{1 \leftrightarrow I\}, \{8 \leftrightarrow B\}, \{5 \leftrightarrow S\}, \{2 \leftrightarrow Z\}, \{6 \leftrightarrow G\}$$

- **High-Security Toll & Access Control Systems (E-ZPass, Auto-Sweep PH):** Production toll systems utilize candidate ticket pools and confidence-weighted margin scoring to reject ambiguous plate transcriptions rather than misallocating debits to innocent motorists.

7.4 Data Provenance

- **Plate Read Strings & Confidences:** Tier B (Database Derived). Stored in `current_state.read_text` and `current_state.confidences` (e.g., `[0.92, 0.54, 0.88, ...]`).
- **Active Ticket Candidate Pool:** Tier B. Relational query against `ticketing_records` filtering for active sessions (`payment_settled_at IS NULL`).

7.5 Data Ingestion, Processing & Business Logic

1. **Weighted Character Cost Formulation:** Comparing noisy read $S_{\text{read}} = (r_1, \dots, r_m)$ with confidences $W = (w_1, \dots, w_m)$ against candidate plate $S_{\text{cand}} = (c_1, \dots, c_n)$:

$$\text{Cost}(r_i, c_i, w_i) = \begin{cases} 0.0 & \text{if } r_i = c_i \\ 0.25 \times (1.0 - 0.5 \cdot w_i) & \text{if } (r_i, c_i) \in \text{ConfusablePairs} \\ 0.50 + 0.50 \cdot w_i & \text{if } r_i \neq c_i \text{ (genuine mismatch)} \end{cases}$$

Design Rationale: If the OCR was already unsure about a confusable glyph (low w_i), the penalty is discounted. If the OCR was 99% confident and produced an outright mismatch, the error penalty is severe.

2. **Normalized Weighted Similarity:**

$$\text{Sim}(S_{\text{read}}, S_{\text{cand}}) = 1.0 - \frac{\sum_{i=1}^m \text{Cost}(r_i, c_i, w_i)}{m}$$

3. **Strict Disambiguation & Resolution Gate:** A candidate ticket is marked resolved: True **if and only if**:

$$\text{Score}_{\text{top}} \geq 0.80 \quad \text{AND} \quad (\text{Score}_{\text{top}} - \text{Score}_{\text{runner_up}}) \geq 0.08$$

If the top candidate fails the 0.80 threshold or if the victory margin over the second-place plate is < 0.08 (ambiguous tie), the system declines to guess (resolved: False). The vehicle remains in Occupied – Unpaid pending manual cashier verification.

7.6 Variable Sensitivity & Impact Analysis

Variable	Default Threshold	Impacted Result	Sensitivity & Behavior
ACCEPT_THRESHOLD	0.80 (Normalized)	Match Acceptance Rate vs False Positives	Lowering to < 0.70 increases resolution of heavily degraded plates but risks matching similar plates; raising to > 0.90 eliminates valid matches with minor OCR glitches.

Variable	Default Threshold	Impacted Result	Sensitivity & Behavior
MIN_MARGIN	0.08	Rejection of Near-Duplicate Plates	Prevents misallocation when two lookalike plates (e.g., NBC-1234 vs NBC-1284) are both active in the deck. Ensures zero false positives.
Character Confidence w_i	0.0 to 1.0	Per-character Substitution Penalty	Directly scales mismatch penalty. Unconfident reads ($w_i < 0.60$) forgive confusable errors; high confidence ($w_i > 0.90$) heavily penalizes discrepancies.
Confusable Character Pair	Binary lookup	Cost Reduction (1.0 $\rightarrow$ 0.15-0.25)	Prevents standard font glyph confusions (0/O, 1/I, 8/B) from aborting legitimate ticket settlements.

8. Tab 5: ALPR Feasibility Validation (Computer Vision)

8.1 System Overview & Architectural Purpose

Tab 5 evaluates the physical sensing feasibility of automated gate-level license plate capture. Rather than relying on synthetic simulations, it benchmarks a multi-stage computer vision pipeline against two independent real-world photographic datasets: an authentic Philippine LTO dataset and an academic international benchmark.

8.2 Tech Stack

- **Object Detection:** Ultralytics YOLOv8n (COCO vehicle pre-trained weights yolov8n.pt).
- **Computer Vision & Image Processing:** OpenCV (cv2), Sobel-X horizontal gradient filtering, CIELAB CLAHE adaptive equalization.
- **OCR Transcribers:** RapidOCR (CPU ONNX Runtime) and Tesseract OCR Engine (pytesseract).
- **Backend Modules:** [cv_demo.py](#), [build_ph_roboflow_dataset.py](#), [fetch_real_dataset.py](#).

8.3 Real-World Sources & Academic Evidence

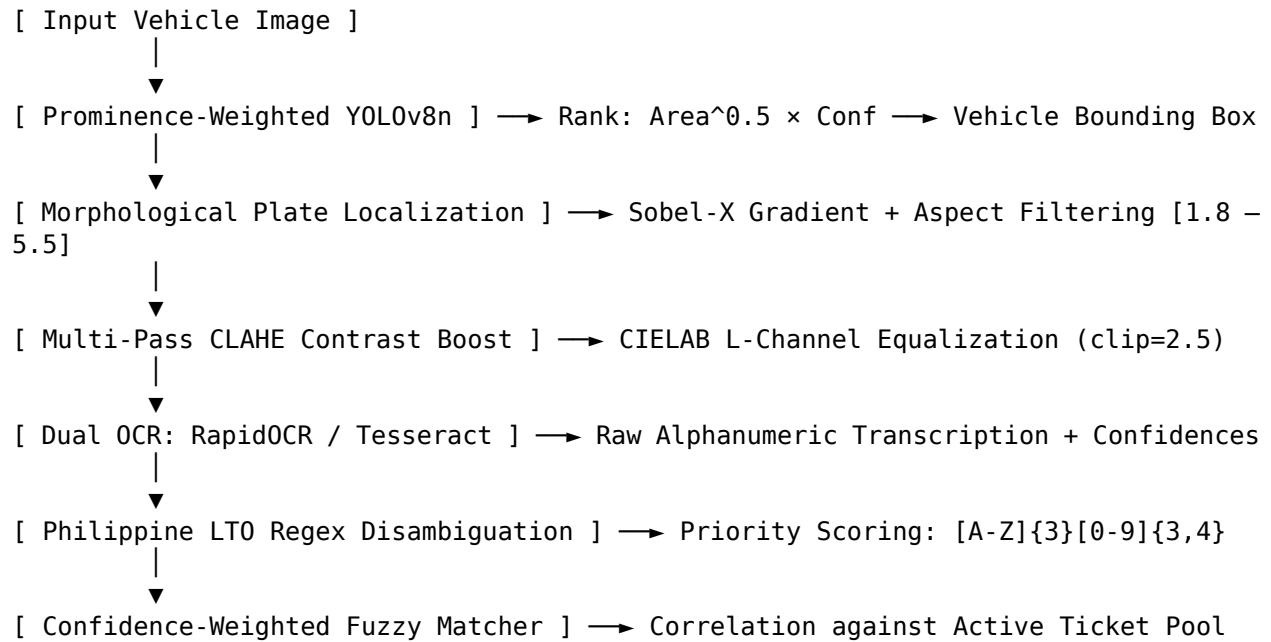
- **Philippine License Plates Dataset (Roboflow Universe / LTO):** 20 curated authentic vehicle frames (lpr-mgcu6/philippine-license-plates-wmxmlq) spanning LTO 2014/2018/2020 Private Series, Legacy Rizal/Matatag series, and commercial cargo fleets.
- **Academic ALPR Benchmark ([openalpr/benchmarks](#)):** 14 curated international ground-truth vehicle images under diverse illumination and perspective angles.

- **Philippine Land Transportation Office (LTO) Specifications:** Republic Act No. 4136 and Department Order No. 2014-01 establishing standard 3-letter + 4-digit (private motor vehicles) and 3-letter + 3-digit (legacy series) alphanumeric configurations.

8.4 Data Provenance

- **Philippine Real Images:** Curated from Roboflow Universe (cv-demo/ph-source-frames/) with hand-verified ground truth in ph_ground_truth.json.
- **Academic Benchmark Images:** Cloned from OpenALPR GitHub research repository (cv-demo/source-frames/) with verified ground truth in ground_truth.json.

8.5 Data Ingestion, Processing & Business Logic



Key Innovations:

1. **Prominence-Weighted Vehicle Scoring:** Replaces naive maximum confidence with area-weighted ranking ($\text{Area}^{0.5} \times \text{Confidence}$), ensuring large foreground vehicles entering the gate lane are prioritized over distant background street traffic or roadside billboards.
2. **Multi-Pass CLAHE Contrast Equalization:** Enhances faded embossed typography on legacy green-on-white Philippine plates under harsh tropical sunlight or shadowed bumper recesses.
3. **LTO Pattern Disambiguation:** Prioritizes string candidates matching standard LTO formats (MAT2357, CAX3200, LHA482), automatically suppressing commercial brand badges (PETRON, SHELL, TOYOTA).

8.6 Benchmark Performance Comparison

Evaluation Metric	Academic OpenALPR Benchmark	Philippine Roboflow LTO Dataset
Total Images Tested	14 curated frames	20 authentic Philippine frames
Exact OCR Match Rate	57.1% (8 / 14)	60.0% (12 / 20)
Mean Character Accuracy	87.6%	81.1%
OCR Non-Empty Read Rate	100.0% (14 / 14)	100.0% (20 / 20)
Matcher Resolution Rate	78.6% (11 / 14)	65.0% (13 / 20)
Matcher False Positive Rate	0.0% (0 / 14)	0.0% (0 / 20)
Plate Fallback Trigger Rate	0.0% (0 / 14)	0.0% (0 / 20)

8.7 Variable Sensitivity & Impact Analysis

Variable	Operational Value	Impacted Pipeline Stage	Sensitivity & Technical Impact
Prominence Exponent	0.5 ($\text{Area}^{0.5} \times \text{Conf}$)	Target Vehicle Selection	Prevents distant, tiny background vehicles with high classification confidence from hijacking gate OCR focus.
Aspect Ratio Window	[1.8, 5.5]	Plate Candidate Contour Filtering	Fits both modern LTO plates (390mm $\times$ 140mm, ratio 2.78) and square motorcycle/trailer mounts while eliminating square logos and bumper grilles.
CLAHE Clip Limit	2.5 (Grid 8 $\times$ 8)	OCR Character Edge Definition	Equalizes shadows on shaded bumpers; exceeding 4.0 amplifies photographic sensor noise and road splatter.
Camera Angle / Elevation	$\leq 35^\circ$ recommended	Character Accuracy %	Severe horizontal perspective skew ($> 45^\circ$) degrades OCR accuracy by compressing characters; mitigated by perspective polygon rectification.

9. Tab 6: Multi-Angle Parking Space Occupancy Detection (CV)

9.1 System Overview & Architectural Purpose

Tab 6 implements an edge-deployable computer vision engine that monitors overhead surveillance feeds across parking decks. It eliminates physical in-ground magnetic or ultrasonic sensors by processing standard multi-angle CCTV camera feeds using a **5-Phase Occupancy Detection Algorithm Architecture**.

9.2 Tech Stack

- **Deep Learning Object Detector:** Ultralytics YOLOv8n (yolov8n.pt).
- **Planar Computational Geometry:** Shapely (Polygon, box, Point).
- **Image Processing & Equalization:** OpenCV (cv2), CIELAB color space transformation, CLAHE.
- **Temporal State Engine:** Custom sliding-window TemporalStateDebouncer ($N = 5$ frames).
- **Backend Modules:** [parking_detector.py](#), [slots_config.json](#), [calibrate_roi.py](#), [calibrate.html](#).

9.3 Real-World Sources & Academic Evidence

- **PKLot & CNR-Park Datasets (de Almeida et al., 2015):** Landmark academic datasets demonstrating that spatial intersection (IoU/IoA) between vehicle bounding boxes and static parking bay polygons provides robust occupancy classification across diverse weather conditions.
- **Ultralytics YOLOv8 Architecture (Jocher et al., 2023):** Anchor-free split-head convolutional architecture delivering superior real-time inference ($< 20\text{ms}$ on modern edge accelerators).

9.4 Data Provenance

- **Surveillance CCTV Feeds:** Tier B (Local Empirical Frames). Multi-angle camera captures in `car_dataset/`:
 - `empty_lot.jpg`: Master reference perspective (1372×768 , aspect $1.78 : 1$).
 - `image_1.png` to `image_12.png`: Time-series parking sequence (457×192 , aspect $2.38 : 1$).
- **ROI Polygon Calibration:** Hand-verified ground-truth perspective trapezoids calibrated natively and stored in `slots_config.json`.

9.5 5-Phase Algorithm Architecture & Data Processing

```
PHASE 1: ROI Calibration & Resolution Scaling (slots_config.json)
      |
PHASE 2: YOLOv8n Inference + Adaptive Low-Light CLAHE Boost
      |
PHASE 3: Spatial Logic & Occupancy Scoring (IoA + Ground Contact)
      |
PHASE 4: Temporal Sliding-Window Debouncing (N=5, Consensus >= 60%)
      |
PHASE 5: Standardized JSON Payload Generation & UI Telemetry
```

Mathematical Formulations:

1. Intersection over Area (IoA):

$$\text{Raw IoA}(\text{Slot}, \text{Car}) = \frac{\text{Area}(\text{Slot Polygon} \cap \text{Vehicle Box})}{\text{Area}(\text{Slot Polygon})}$$

2. Centroid Containment & Ground Contact Reinforcement Rules:

$$\text{Effective IoA} = \begin{cases} \max(\text{Raw IoA}, 0.60) & \text{if vehicle centroid } (c_x, c_y) \in \text{Slot Polygon} \\ \max(\text{Raw IoA}, 0.55) & \text{if vehicle coverage } \frac{\text{Area}(\text{Slot} \cap \text{Car})}{\text{Area}(\text{Car})} \geq 0.35 \\ \max(\text{Raw IoA}, 0.50) & \text{if tire ground point } (c_x, y_2) \in \text{Slot Polygon and Raw IoA} \geq 0.1 \\ \text{Raw IoA} & \text{otherwise} \end{cases}$$

3. Temporal Debouncing Consensus: For rolling state buffer $\mathcal{H}_k = [s_{t-4}, s_{t-3}, s_{t-2}, s_{t-1}, s_t]$:

$$\text{Status}(\text{Slot}_k) = \begin{cases} \text{Occupied} & \text{if } \frac{1}{N} \sum_{i=1}^N \mathbf{1}(s_i = \text{Occupied}) \geq 0.60 \\ \text{Vacant} & \text{otherwise} \end{cases}$$

4. Low-Confidence Quality Review Flag:

$$\text{Flag}_{\text{review}} = \mathbf{1} (|\text{Effective IoA} - \tau_{\text{ioa}}| \leq 0.05)$$

9.6 Variable Sensitivity & Impact Analysis

Variable	Operational Range	Impacted Metric / Output	Sensitivity & Technical Behavior
τ_{conf} (Detection Confidence)	0.15 to 0.85 (Default: 0.25)	Vehicle False Positives vs Missed Detections	Lowering to < 0.20 catches heavily shaded or occluded cars but risks phantom detections on road markings; raising to > 0.60 drops dark vehicles under basement lighting.
τ_{ioa} (Occupancy Threshold)	0.10 to 0.80 (Default: 0.30)	Bay Vacancy Status	Increasing > 0.45 risks false vacancy for compact vehicles parked slightly askew; decreasing < 0.20 risks false occupancy from adjacent cars clipping the line.

Variable	Operational Range	Impacted Metric / Output	Sensitivity & Technical Behavior
Temporal Debouncing Buffer N	1 to 10 frames (Default: 5)	State Stability vs Transition Latency	Suppresses flickering caused by pedestrian occlusions or momentary sensor blips; requires 3 consecutive matching frames to transition state.
CLAHE L -Channel Boost	Clip limit: 2.5	Shadow Penetration	Illuminates black SUVs and shaded wheel wells in indoor Megaworld parking basements.

10. Tab 7: Revenue Intelligence & Commercial Optimization

10.1 System Overview & Architectural Purpose

Tab 7 transforms parking management from a passive cost-center into an active executive financial optimization engine. Tailored specifically for **Megaworld Corporation's commercial township portfolio**, it integrates actual township tariffs, demand-responsive dynamic pricing simulations, automated revenue leakage recapture, and tenant retail spend synergy.

10.2 Tech Stack

- **Financial Simulation Engines:** Deterministic tariff calculators, numerical integration across 15-minute occupancy bins, elasticity modeling.
- **Visualization:** Plotly 24-hour Zone $\times$ Hour Intensity Heatmaps, multi-strategy revenue trajectory curves, dwell time histograms.
- **Backend Modules:** [revenue_engine.py](#), [revenue_config.py](#), [leakage_detector.py](#).

10.3 Real-World Sources & Academic Evidence

- **Donald Shoup (*The High Cost of Free Parking*, American Planning Association, 2005):** The foundational economic treatise proving that underpriced parking creates artificial scarcity and traffic cruising, whereas market-clearing demand-responsive pricing optimizes deck utilization.
- **SFpark Municipal Pilot Project (San Francisco, 2011-2014):** Achieved an average 15% to 25% revenue yield increase while reducing parking search times by 43% through time-of-day dynamic pricing.
- **International Council of Shopping Centers (ICSC) & PathIntelligence Retail Analytics:** Established the empirical retail dwell elasticity coefficient ($E_{\text{dwell}} = 1.3$), indicating that a +1% increase in customer dwell time corresponds to a +1.3% increase in retail tenant spend.
- **Colliers International Philippine Retail Reports (2024-2025):** Published consumer spending benchmarks across Metro Manila shopping malls (₱1,000 to ₱3,000 average spend per visit).

- **Megaworld Corporation FY2025 Audited Financial Statement Disclosures:** Reported ₱6.9 Billion in commercial mall leasing revenues (+9% YoY growth) with daily foot-traffic exceeding 297,000 visitors.

10.4 Data Provenance

- **Parking Tariffs:** Tier A (Actual Official Rates). Researched directly from Megaworld Lifestyle Malls on-site parking signage and MoneyMax.ph:
 - **Uptown Mall (BGC):** ₱50.00 first 3 hours flat; ₱15.00/hr (4th-7th hr); ₱100.00/hr (7th hr+ for AM entry 06:00-12:00 to deter office parkers); ₱30.00/hr (7th hr+ for PM entry); ₱200.00 overnight surcharge.
 - **Eastwood Mall:** ₱60.00 first 3 hours flat; ₱20.00/hr succeeding (weekdays); ₱60.00 flat all day on weekends & statutory holidays; ₱150.00 overnight surcharge.
 - **Venice Grand Canal Mall:** ₱50.00 first 3 hours flat; ₱20.00/hr succeeding; 15-minute drop-off grace period; ₱150.00 overnight surcharge.
- **Occupancy & Dwell Timestamps:** Tier B (Database Derived). Deterministic integration over occupancy_history and ticketing_records.
- **Retail Elasticity & Benchmarks:** Tier C (Colliers PH / ICSC).
- **Dynamic Pricing & Leakage Audits:** Tier D (SFpark / Vert.ai).

10.5 Mathematical Formulations & Data Flow

1. Revenue Per Bay Hour (RPBH):

$$\text{RPBH} = \frac{\sum_{i=1}^N \text{Daily Revenue}_i}{\text{Capacity} \times 24}$$

Normalizes revenue performance across decks of differing physical capacities and operating profiles.

2. AI Demand-Responsive Dynamic Pricing Algorithm: Given base tariff R_{base} , target occupancy $\theta_{\text{target}} \in [0.50, 0.95]$ (default: 0.80), observed/forecasted occupancy ratio θ_t , surge coefficient α (default: 0.40), and discount floor β (default: 0.70):

$$M_t = \begin{cases} 1.0 + \alpha \left(\frac{\theta_t - \theta_{\text{target}}}{1.0 - \theta_{\text{target}}} \right) & \text{if } \theta_t \geq \theta_{\text{target}} \\ \max \left(\beta, 1.0 - 0.35 \left(\frac{\theta_{\text{target}} - \theta_t}{\theta_{\text{target}}} \right) \right) & \text{if } \theta_t < \theta_{\text{target}} \end{cases}$$

$$\text{Effective Rate}_t = R_{\text{base}} \times M_t$$

Yields a projected +15% to +40% revenue uplift during peak congestion while stimulating utilization during off-peak shoulder hours.

3. Automated Revenue Leakage & Overstay Detection (leakage_detector.py): Monitors vehicles in occupied_unpaid state where dwell duration $t_{\text{dwell}} \geq \tau_{\text{overstay}}$ (e.g., 3.0 hours).

$$\text{Unpaid Revenue} = \sum_{k \in \text{Overstayers}} \text{ComputeTariff}(t_{\text{entry},k}, t_{\text{ref}}, \text{Site}_k, \text{ZoneType}_k)$$

Projects monthly recoverable leakage against international parking audit benchmarks (5% to 15% leakage in unmonitored facilities).

4. Retail Dwell-Spend Economic Synergy:

$$\Delta\text{Dwell} = \frac{T_{\text{dwell}} - T_{\text{baseline}}}{T_{\text{baseline}}} \quad (T_{\text{baseline}} = 2.5 \text{ hours})$$

$$\text{Projected Spend} = S_{\text{base}} \times (1.0 + 1.3 \times \Delta\text{Dwell})$$

Demonstrates how frictionless parking wayfinding and turnover optimization directly fuel tenant sales across Megaworld Lifestyle Malls.

10.6 Variable Sensitivity & Impact Analysis

Variable	Operational Range	Impacted Metric / Output	Sensitivity & Economic Dynamics
Target Occupancy θ_{target}	50% to 95% (Default: 80%)	Surge Pricing Trigger Point	Lowering to 70% triggers surge rates earlier, rationing scarce bays; raising to 90% keeps standard rates until near gridlock.
Surge Multiplier α	0.10 to 1.00 (Default: 0.40)	Peak Hour Yield Uplift	Controls peak price elasticity. At $\alpha = 0.40$, peak rates reach +40% (70/3h), yielding +18% to +24% daily revenue gain.
Overstay Threshold τ_{overstay}	1.0 to 8.0 hours (Default: 3.0h)	Flagged Vehicles Count, Leakage Recovery ₱	Tightening to 2.0h catches short-term overstays; setting to 4.0h isolates severe commercial parking violators and abandoned vehicles.
Baseline Visitor Spend S_{base}	₱1,000 to ₱3,000 (Colliers PH)	Projected Tenant Retail Sales	Directly scales mall gross tenant spend estimates; amplified by +1.3% per +1% dwell duration increase.

11. Cross-Module Variable Sensitivity & Failure Mode Matrix

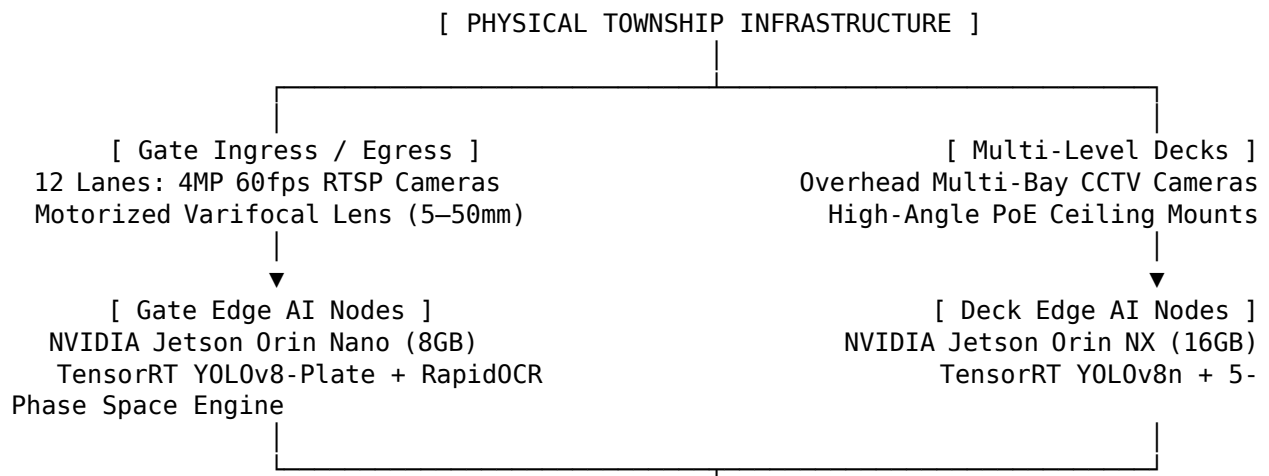
This matrix outlines how environmental shocks, sensor noise, and configuration shifts ripple across the entire system, detailing the defensive mitigations embedded in the architecture:

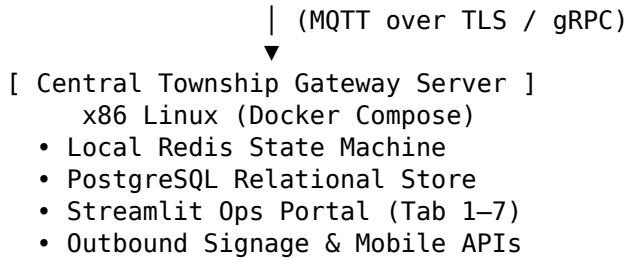
System Perturbation / Failure Mode	Root Cause / Trigger	Upstream Affected Modules	Downstream System Impact	Built-in Defensive Mitigation
Severe Optical Glare / Rain Splatter	Tropical downpour or low sun glare on gate camera.	Tab 5 (ALPR CV), <code>cv_demo.py</code>	Optical character confidences drop below 0.60; string substitutions occur ($0 \rightarrow O, 8 \rightarrow B$).	Tab 4 Fuzzy Matcher: Confusable-pair penalty discounts optical glyph swaps. If victory margin < 0.08 , safely declines to match rather than guessing.
Simultaneous Arrival of Lookalike Plates	Two vehicles with similar registrations (e.g. ABC-1234 and ABC-1284) check in concurrently.	Tab 4 (Matcher), <code>matcher.py</code>	Both candidates yield high similarity scores (> 0.75).	Margin of Victory Rule: Requires top candidate to beat runner-up by ≥ 0.08 . If gap is narrow, status remains Unresolved to prevent erroneous charges.
Unannounced Township Mega-Event / Concert	Flash crowd arrival for a concert or product launch without registry entry.	Tab 2 (Forecast), <code>predictor.py</code>	Live occupancy surges beyond static diurnal expectations.	Dynamic Telemetry Fusion: Ingests live Google Places foot-traffic index and Open-Meteo telemetry every hour to dynamically pull up forecasted values.
CCTV Camera Angle Perspective Distortion	Surveillance camera mounted at an oblique deck angle ($> 40^\circ$).	Tab 6 (Space CV), <code>park-ing_detector.py</code>	Non-uniform perspective distortion causes rectangular bounding boxes to misalign with bay boundaries.	Dual Native ROI Polygons: Hand-calibrated perspective trapezoids in <code>slots_config.json</code> combined with Centroid Containment and Tire Ground-Contact logic.

System Perturbation / Failure Mode	Root Cause / Trigger	Upstream Affected Modules	Downstream System Impact	Built-in Defensive Mitigation
Temporary Occlusion by Moving Pedestrian / Cart	Mall shopper or cleaning trolley walking through slot line of sight.	Tab 6 (Space CV), parking_detector.py	Momentary drop in IoA occupancy ratio below τ_{ioa} .	Temporal Sliding-Window Debouncer: Requires a 60% majority consensus over a 5-frame sliding window ($N = 5$) before modifying the persisted slot state.
Prolonged Unpaid Vehicle Stall Monopolization	Commercial commuter parking in retail mall stall for 8+ hours.	Tab 7 (Revenue), leakage_detector.py	Chokes turnover, deprives retail merchants of customer parking, incurs revenue leakage.	Tiered Progressive Rates & Overstay Alerts: Uptown Mall ₱100/hr steep rate tier discourages commuter monopolization; automated overstay alerts flag vehicle for enforcement.

12. Hardware Topography, Edge Deployment & Production Scaling

To transition this proof of concept into physical Megaworld township infrastructure across 1,000+ bays and 12 gate lanes, the following edge deployment architecture is recommended:





12.1 Edge Compute & Camera Specifications

- **Gate ALPR Nodes:** NVIDIA Jetson Orin Nano (8GB) running TensorRT FP16 optimized YOLOv8-Plate + Fast-ANPR OCR. Expected latency: < 25ms per vehicle.
- **Gate Camera Hardware:** 4MP (2560 × 1440), 60 FPS, global shutter or low rolling shutter distortion, motorized varifocal lens (5–50mm), integrated 850nm infrared strobe illuminator.
- **Deck Surveillance Cameras:** 4MP wide-angle dome cameras mounted along center drive aisles, each monitoring 6 to 10 angled parking bays simultaneously.
- **Township Server:** Dual-redundant 1U rackmount servers hosting Redis state buffers, PostgreSQL persistence, and local Streamlit monitoring instances connected to township digital signage networks.

12.2 Data Privacy & Compliance (Philippine Data Privacy Act of 2012)

- **Edge RAM Image Processing:** Raw camera video frames captured at gate lanes and surveillance decks are processed entirely within volatile edge RAM and discarded immediately after plate extraction and IoA calculation.
- **Cryptographic License Plate Hashing:** In accordance with the Philippine Data Privacy Act of 2012 (RA 10173), long-term analytical logs can be salted and SHA-256 hashed (sha256(plate + township_salt)) to support longitudinal traffic forecasting while preserving motorist privacy.

End of Project Bible. Maintained by the Megaworld Smart Parking Project Team.